

Understanding the black box: How Neural Networks Extract Cosmological Information from the Cosmic Web

When a black box meets a black box | *which parts of the cosmic web is a CNN actually looking at?*

AUTHORS

Arnab Lahiry • Adrian E. Bayer • Francisco Villaescusa-Navarro

PUBLISHED

ApJ 994, 129 • Nov 20 2025

ARXIV

2504.17839v2

PRESENTED BY

Arnab Lahiry

“What happens when a *black box* — a *neural network* —
meets a *black box* — a *simulation of the Universe*?”



Why throw machine learning at cosmology?

Upcoming surveys — **Euclid**, **Rubin/LSST**, **DESI**, **SKA** — will deliver matter maps at unprecedented resolution. The signal lives on small scales where:

The matter distribution is highly **non-Gaussian** — power spectra no longer capture all the information.

Baryonic physics (SN feedback, AGN, winds) contaminates exactly the regime that carries the most modes.

Analytic summary statistics aren't theoretically guaranteed to be optimal.

LATE-TIME OBSERVABLES

Galaxy clustering Weak lensing 21 cm
intensity mapping CMB secondaries (SZ)

THE HOPE

*Field-level inference: every pixel, no
compression.*

From a Gaussian CMB to a *non-Gaussian* cosmic web.

$z \approx 1100$ — LAST SCATTERING

CMB temperature fluctuations are very nearly Gaussian. The power spectrum *completely* characterises the field.

$P(k)$ suffices · two-point $\equiv$ all-point

$z \approx 0$ — TODAY

Gravity has folded matter into halos, filaments, walls and voids. The field is **highly non-Gaussian** — power spectrum is no longer sufficient.

$P(k) \subsetneq$ information · need higher moments OR field-level

THE BIND

Where the signal is richest (small scales), the modeling is hardest (baryons).

Field-level inference, briefly.

Rather than compress a density field down to a summary statistic, feed every pixel into the inference pipeline. A growing literature explores this — *Jasche & Wandelt 2013, Seljak+ 2017, Schmidt+ 2019, Dai & Seljak 2022, Bayer+ 2023, Lemos+ 2024*, and many more.

THE LAUNCHPAD FOR THIS PAPER

Villaescusa-Navarro et al. 2021b — trained a CNN on 2D projected total-matter maps from CAMELS hydrodynamic simulations. Recovered Ω_m and σ_8 with $R^2 \approx 1$ even while marginalising over four feedback parameters.

BUT ...

CNNs are notoriously opaque. Which features of the cosmic web is it actually using?

THE THESIS OF THE PAPER

Which **Fourier scales** , which density regimes ,
and which morphological features does the
CNN extract information from?

And critically — does the answer change in the presence of complicated baryonic physics?



What we'll cover.

~10 min

§ 01

Setup — CAMELS, the CNN, and three interpretability tools

~5 min

§ 02

Where is the CNN looking? — attribution maps with no cuts

~8 min

§ 03

Fourier-scale cuts — how much do small scales matter?

~15 min

§ 04

Density cuts — voids vs. halos, hydro vs. N-body

~7 min

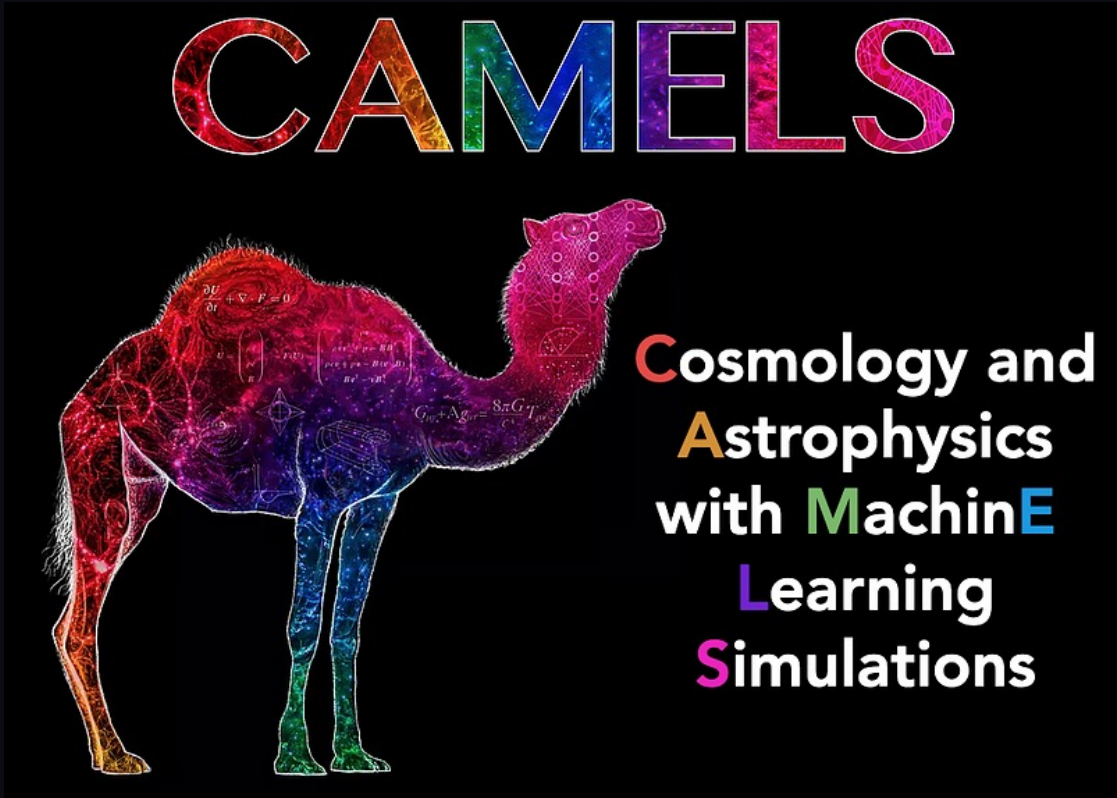
§ 05

Discussion, caveats & implications

01 ----- SETUP

The simulations, the network, the tools.

CAMELS · IllustrisTNG · CNN architecture · Saliency, IG, GradientSHAP



Over **16,000 simulations** spanning a 6-D parameter space.

Both **Hydrodynamic** and **Gravity-only N-body** variants

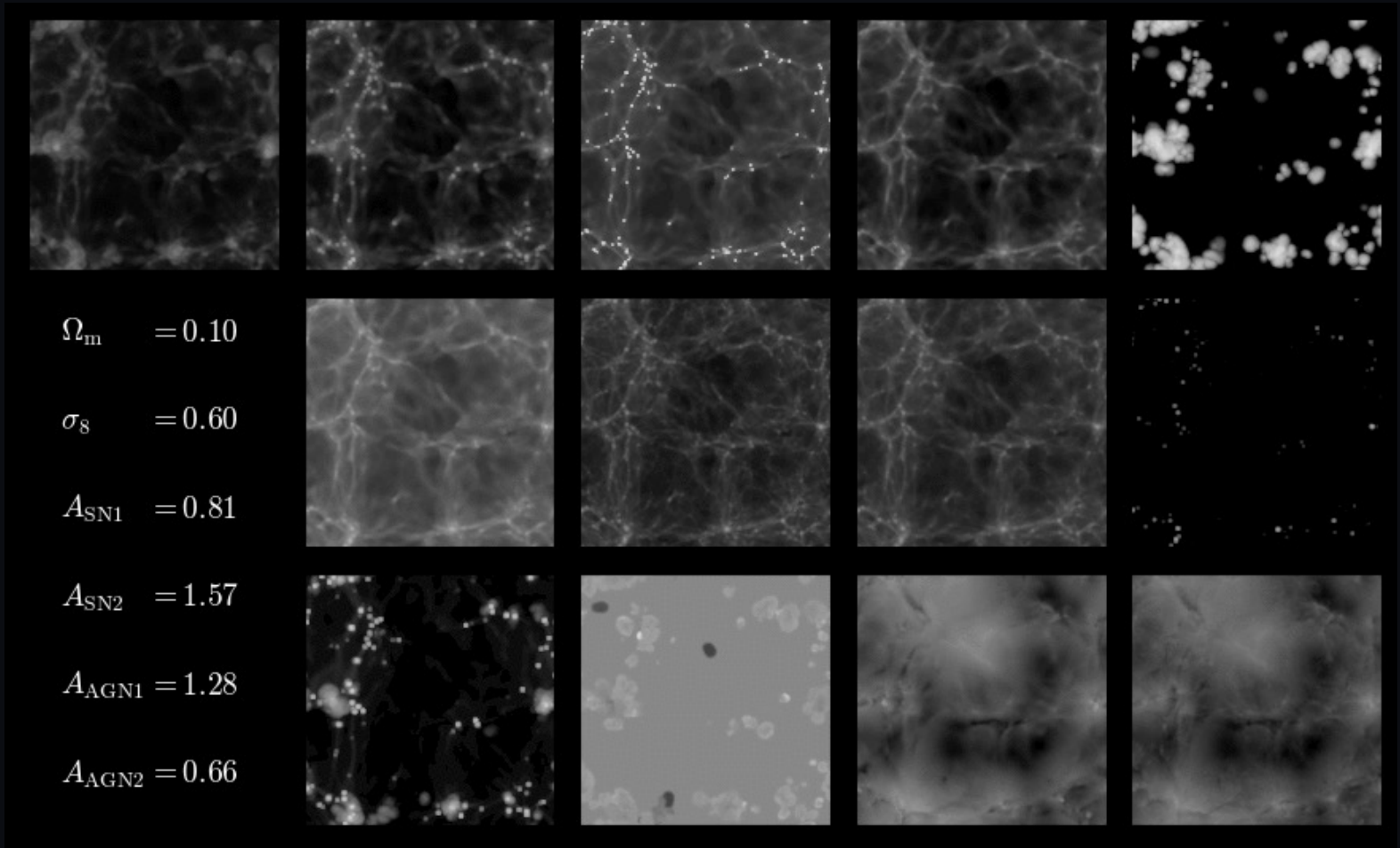
This paper uses the **IllustrisTNG Latin-Hypercube**

— same subgrid physics as IllustrisTNG, run with AREPO.

2×256^3 dark + gas elements · box of $(25 \text{ h}^{-1} \text{ Mpc})^3$ $z = 127 \rightarrow 0$.

Each sim has different cosmology, different feedback, different initial seed.

SIMULATIONS 16k+	PARAMETERS 6
BOX SIZE $25 \text{ h}^{-1} \text{ Mpc}$	MAP RESOLUTION 256^2



Six parameters: *2 cosmological, 4 astrophysical.*

PARAM	DESCRIPTION	RANGE
Ω_m	Total matter density (dark + baryonic) relative to critical density	[0.1 , 0.5]
σ_8	Amplitude of the linear density field on 8 h ⁻¹ Mpc scales at z = 0	[0.6 , 1.0]
A_{SN1}	Galactic winds (supernova feedback) — energy per unit star-formation rate	[0.25 , 4.0]
A_{SN2}	Galactic-wind speed (supernova feedback)	[0.5 , 2.0]
A_{AGN1}	Kinetic-mode AGN feedback — energy per unit BH accretion rate	[0.25 , 4.0]
A_{AGN2}	Kinetic-mode AGN matter-ejection speed	[0.5 , 2.0]

Other cosmological parameters fixed: Ω_b =0.049, h=0.6711, n_s =0.9624, $\sum m_\nu$ =0, w=-1 . Factor-of-4 variation in the feedback knobs makes the marginalisation non-trivial.

A 2D projected total-matter map for each simulation, taken at $z = 0$.

Each simulation snapshot is projected to a single 256×256 image of $M_{\text{tot}}(x, y)$.

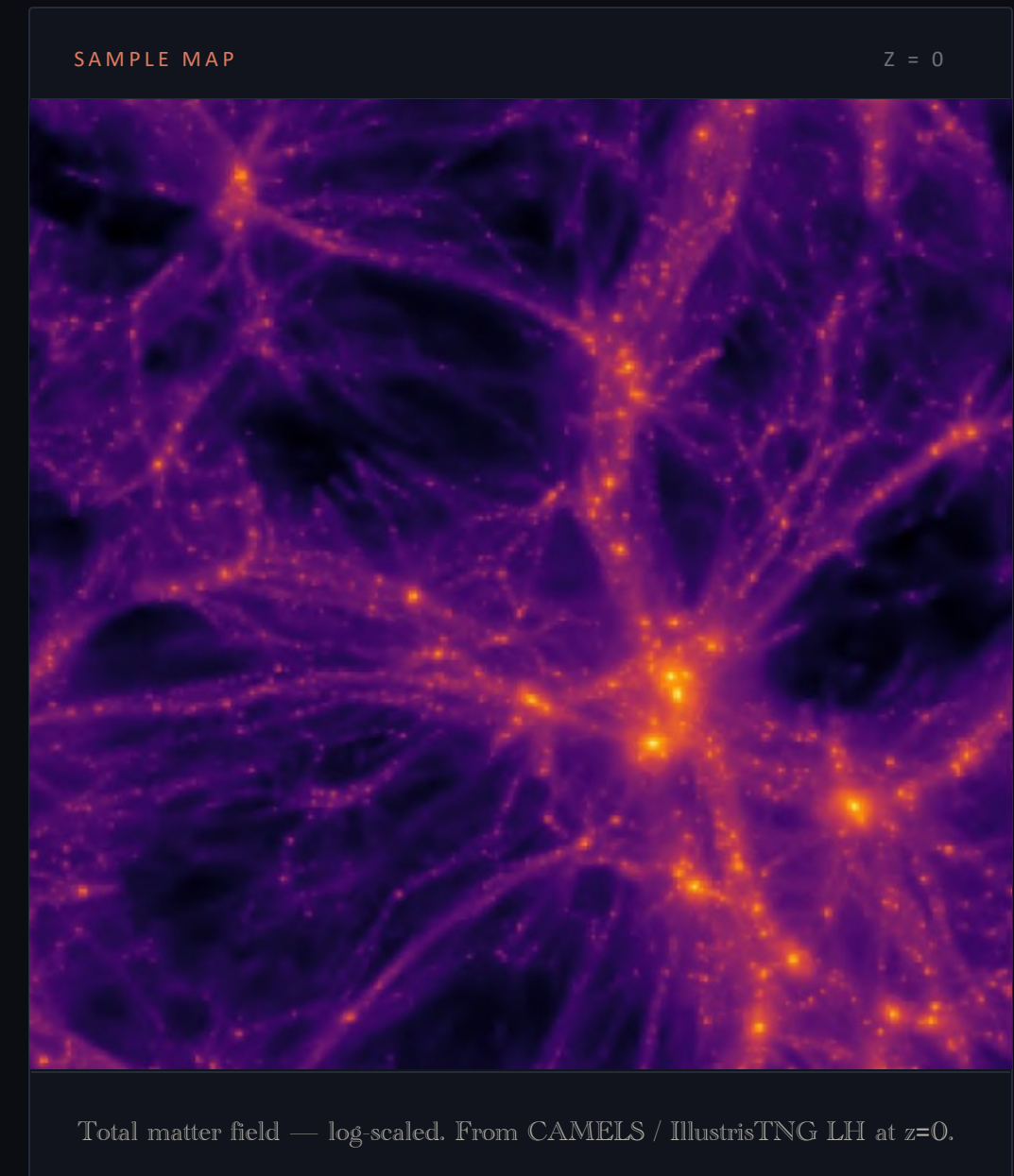
Pixel values span **several orders of magnitude** — voids to halos.

Maps are pulled from the **CAMELS Multifield Dataset (CMD)**.

Bright knots $\rightarrow$ halos · threads $\rightarrow$ filaments · dark patches $\rightarrow$ voids.

SET SIZE

15,000 maps · 90 / 5 / 5 train / validation / test

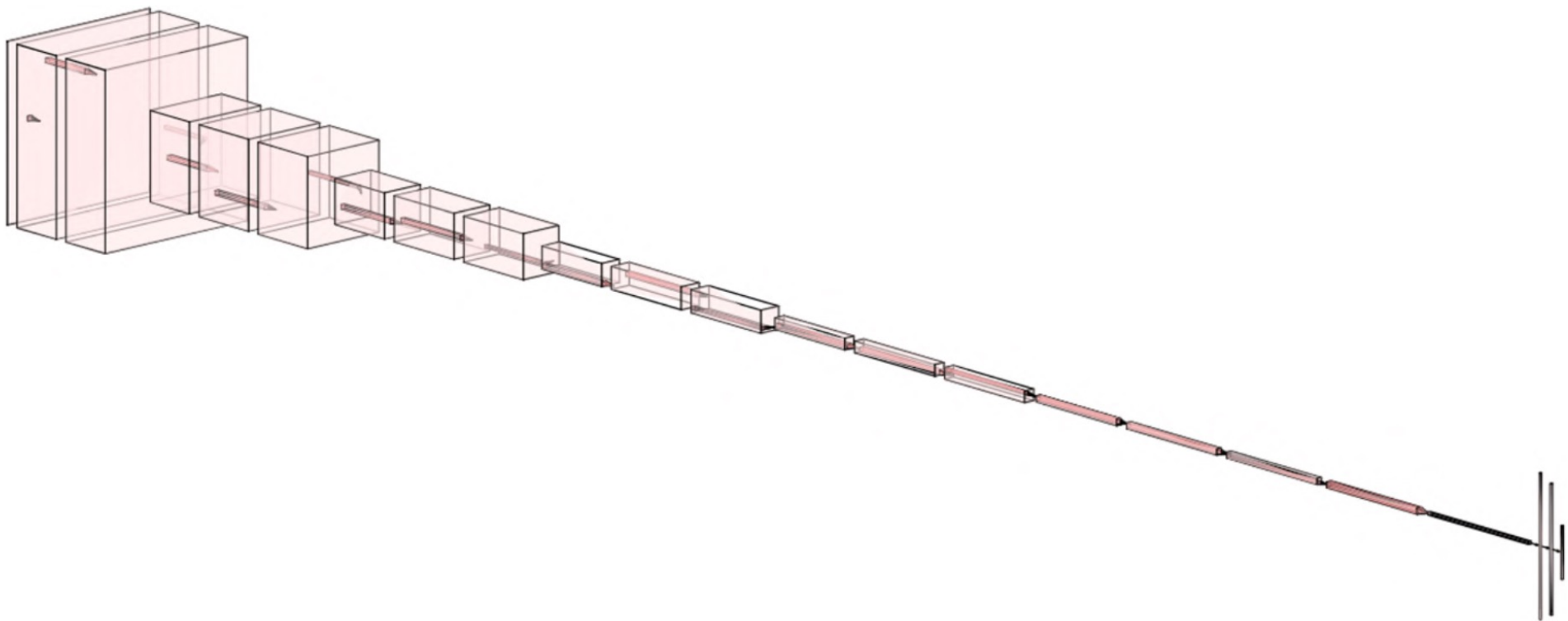


The CNN architecture

Mimics the architecture used in Villaescusa-Navarro+ 2021b

FIGURE 1 — SCHEMATIC

CONV K=4, S=2, P=1 · BN · LEAKYRELU



Stack of convolutional blocks → flattened → two fully-connected layers → 12-element output: posterior mean and variance for each of the six parameters.

CONV STACK

kernel 4 · stride 2 · padding 1 BatchNorm2D +
LeakyReLU after each.

OUTPUT

12-vector: $\langle \mu_i, \sigma_i^2 \rangle$ for $i = 1 \dots 6$.

HYPERPARAMETER SEARCH

Optuna · 50 trials · TPE sampler.
Tunes depth, dropout, weight decay, LR.

Simulation-based inference.

The network outputs the **marginal posterior mean and variance** for each parameter. Predicted moments are matched to the truth via a moment-network loss (*Jeffrey & Wandelt 2020*):

$$\mathcal{L} = \sum_{i=1}^6 \ln \left[\sum_{j \in \text{batch}} (\mu_{ij} - \theta_{ij})^2 \right] + \sum_{i=1}^6 \ln \left[\sum_{j \in \text{batch}} ((\mu_{ij} - \theta_{ij})^2 - \sigma_{ij}^2)^2 \right]$$

left: forces $\mu \rightarrow \text{truth}$ · right: calibrates σ^2 to match the actual squared error

Marginal posterior moments are computed as:

$$\mu_i(X) = \int \theta_i p(\theta_i | X) d\theta_i$$

$$\sigma_i^2(X) = \int (\theta_i - \mu_i(X))^2 p(\theta_i | X) d\theta_i$$

QUALITY METRIC

Coefficient of determination

$$R^2 = 1 - \Sigma(\mu_i - \theta_i)^2 / \Sigma(\theta_i - \bar{\theta})^2$$

$R^2 \approx 1$ means tight prediction · $R^2 \rightarrow 0$ means no skill.

Three attribution methods.

All three are gradient-based attribution methods — they score each input pixel by how much it contributed to a particular output. Implemented via the **Captum** library.

01 -- SALIENCY MAP

Local sensitivity.

Simonyan, Vedaldi & Zisserman 2013

Simply $\partial f(x) / \partial x_i$ — the gradient of the output with respect to each input pixel.

Cheap, intuitive

Noisy

Gradient saturation in deep nets

02 -- INTEGRATED GRADIENTS

Path average.

Sundararajan, Taly & Yan 2017

Integrate the gradient along a straight line from a **baseline** input (zero / noise) to the actual input.

Resolves saturation

Global view

One baseline choice

03 -- GRADIENTSHAP

Shapley-style average.

Lundberg & Lee 2017

Average Integrated Gradients over a **distribution** of baselines. Connects to game-theoretic Shapley values.

Most stable

Captures feature interactions

More expensive

METHOD 01

Saliency

INTUITION

Asks: if I wiggled this pixel infinitesimally, how would the output respond?

$$S_i(x) = \partial f(x) / \partial x_i$$

Large $|S_i(x)| \Rightarrow$ the model is locally sensitive to pixel i . **Two known pathologies:**

Noisy — pixel-level gradients fluctuate wildly.

Gradient saturation — important features can have near-zero gradient in deep nets.

METHOD 02 — THE WORKHORSE OF THIS PAPER

Integrated Gradients.

$$IG_i(x) = (x_i - x'_i) \cdot \int_0^1 \frac{\partial f(x' + \alpha(x - x'))}{\partial x_i} d\alpha$$

Integrate the gradient along the straight line from a baseline x' (often zero or white noise) to the actual input x .

Resolves saturation — sums gradient contributions across the whole path.

More stable + more **global** view of importance

Used for everything from §3.2 onwards.

INTUITION

Asks: *how much does pixel i contribute to $f(x)$, accumulated along a path from "nothing" to "this image"?*

METHOD 03

Gradient *SHAP*.

INTUITION

Asks: *averaging over many "starting points", how much does pixel i matter?*

$$\phi_i(x) = \mathbb{E}_{x' \sim p(x')} \left[(x_i - x'_i) \int_0^1 \frac{\partial f(x' + \alpha(x - x'))}{\partial x_i} d\alpha \right]$$

Like Integrated Gradients but **averaged over a distribution of baselines** . Grounded in cooperative-game Shapley values.

Captures interactions between features.

Most stable of the three.

Computationally most expensive out of the three

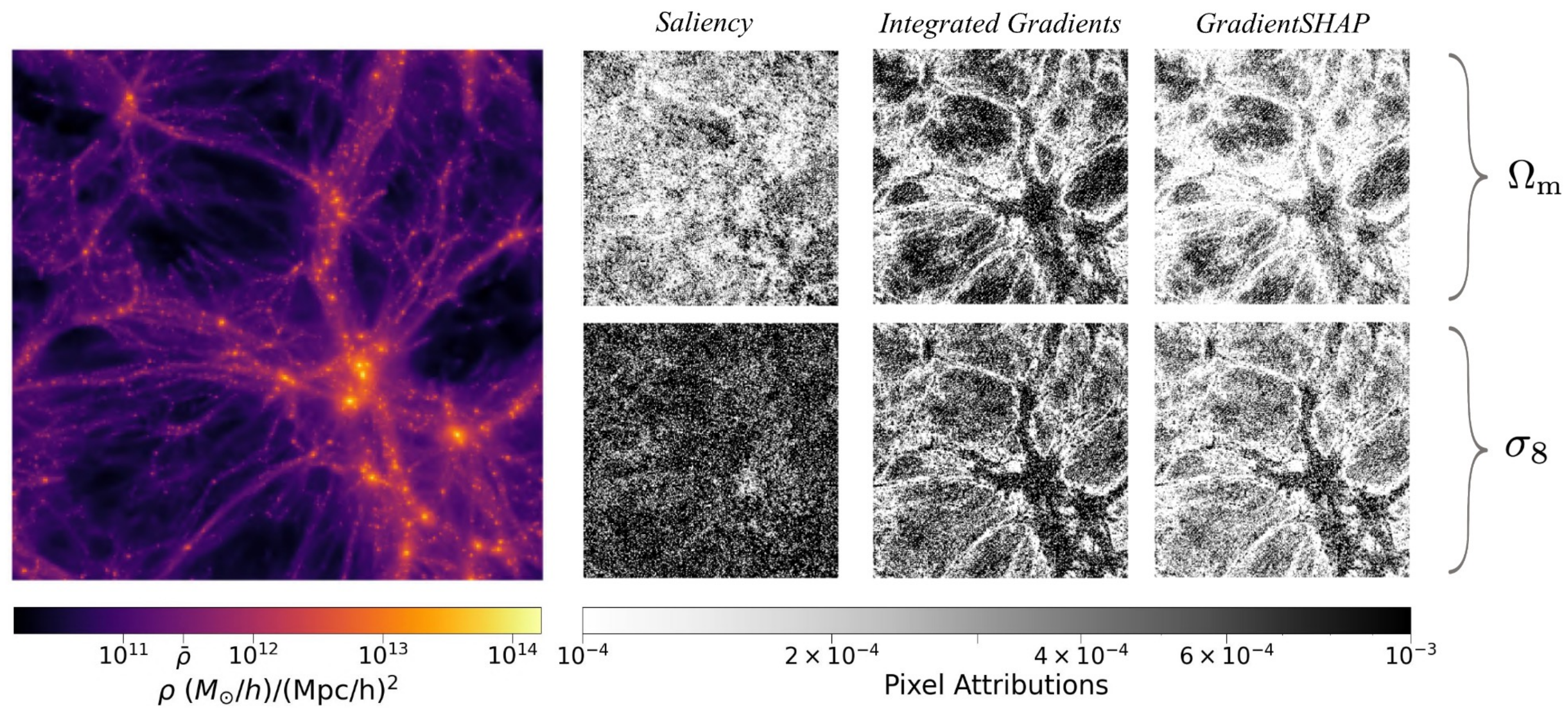
02 ----- WHERE IS THE INFORMATION?

First look: **no cuts** at all.

Apply each of the three **attribution methods** to the raw maps and see **what lights up**.

Attribution maps for the raw cosmic web.

FIGURE 2

COSMIC WEB (LEFT) · ATTRIBUTION MAPS FOR Ω_m (TOP), σ_8 (BOTTOM)

Black = high attribution. Saliency is noisy; Integrated Gradients $\approx$ GradientSHAP. Both show strong weight on deep voids and overdense clusters; thick filaments matter, thin filaments do not.

First takeaway | the CNN attends to *the extremes*.

FOR Ω_M

The CNN focuses on **deep voids** and **clusters/halos** . Thick filaments have meaningful weight; thin ones essentially none. The network is learning from the most **extreme regions** of the field.

FOR σ_8

Same pattern, but with a tilt: **overdense regions slightly outweigh underdense regions** . (Makes sense — σ_8 is an amplitude on small scales.)

METHODOLOGICAL CHOICE FOR THE REST OF THE PAPER

*IG $\approx$ GradientSHAP visually and quantitatively $\rightarrow$ stick with *Integrated Gradients* (cheaper, robust, global).*

The pipeline for every experiment.

For each value of a cut threshold, repeat:

01 -- PREPROCESS Cut the maps Top-hat in k for Fourier cuts; replace below/above-threshold pixels with log-uniform random values for density cuts.	02 -- TRAIN Retrain the CNN Fresh model per cut value. 90/5/5 split. Augment with rotations and flips.	03 -- ATTRIBUTE Integrated Gradients Run IG on the test set to see which pixels the new model relies on.	04 -- EVALUATE R^2 & $\langle \sigma^2 \rangle$ Quantify accuracy and precision. Track how they change as a function of the cut.
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How much do small scales matter?

Top-hat filter in k-space $\rightarrow$ retrain $\rightarrow$ measure R^2 and $\langle \sigma^2 \rangle$ as a function of $k_{\max}$.

A tension on small scales.

SMALL SCALES HELP

Number of modes grows as k_{max}^2 (in 2D). More modes $\Rightarrow$ more independent measurements $\Rightarrow$ tighter constraints.

SMALL SCALES HURT

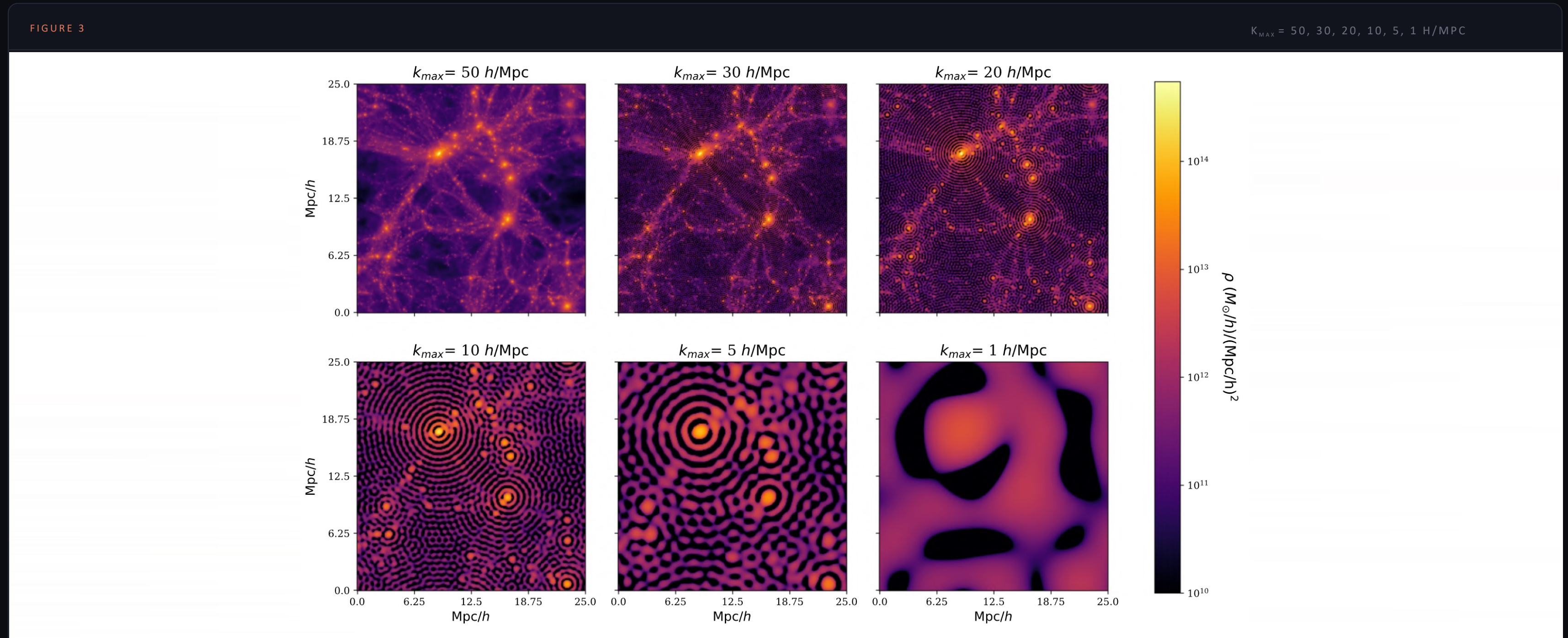
Baryonic effects dominate on small scales — feedback, winds, AGN — and they’re the least well-modeled regime.

EXPERIMENT

Convolve each map with a top-hat- K , -filter (zero out modes above k_{max}); sweep $k_{\text{max}} = 50, 30, 20, 10, 5, 1 \text{ h/Mpc}$; retrain at every step.

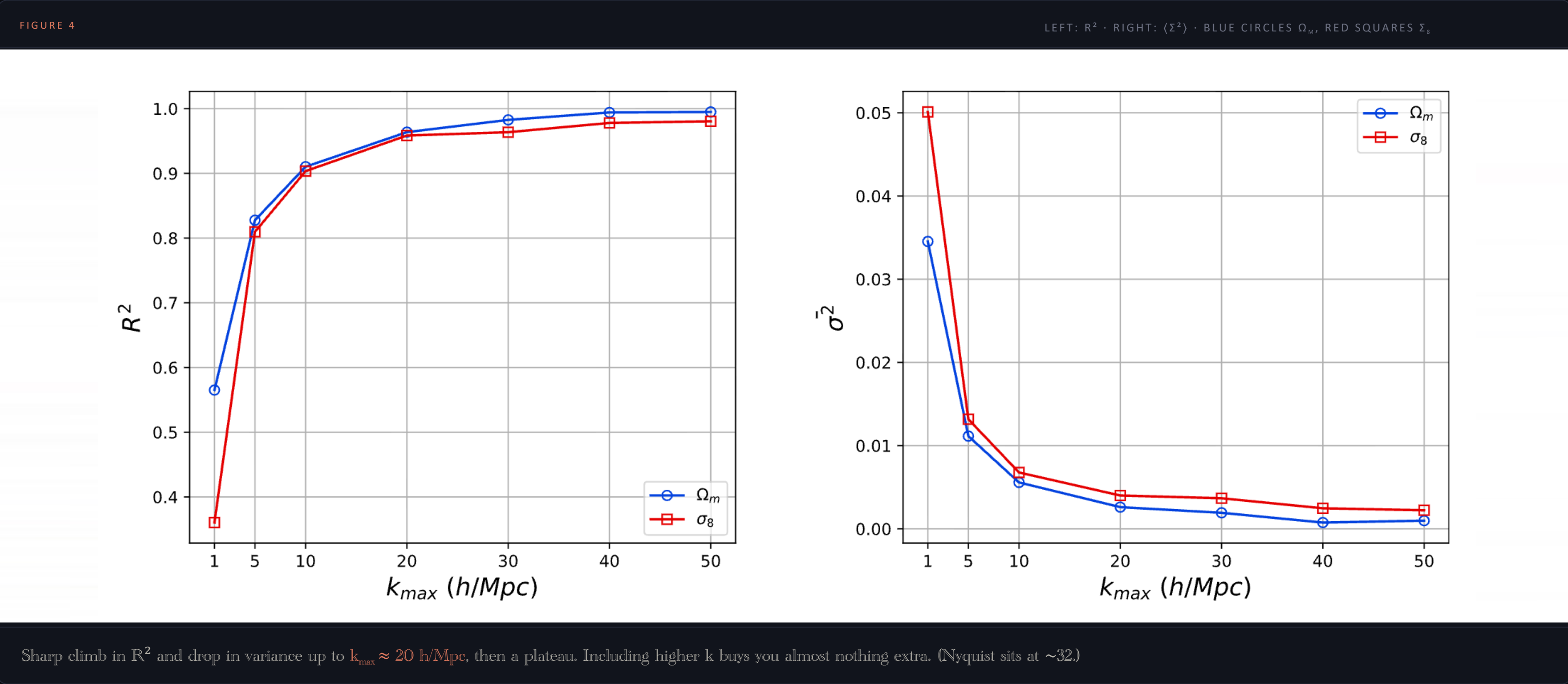
For reference: the Nyquist frequency of the 256^2 maps is $\sim 32 \text{ h/Mpc}$.

Figure 3 — the maps at decreasing $k_{\max}$.



At $k_{\max} = 50$ the map is barely altered. By $k_{\max} = 20$ small clumps soften. By $k_{\max} = 5$ halos and most filaments are gone — only the largest-scale structure remains.

Figure 4 — inference quality vs k_{max}



Aggressive **k**-cuts cost *almost nothing*.

INFORMATION SATURATES BY

$k \approx 20$

h/Mpc — well below the Nyquist limit of ~ 32

R^2 DROP FROM $K=50 \rightarrow K=10$

$\approx 10\%$

For Ω_m and σ_8 alike

IMPLICATION

Cut the regime baryons own.

Field-level inference can be made robust to subgrid uncertainty with minimal cost in precision.

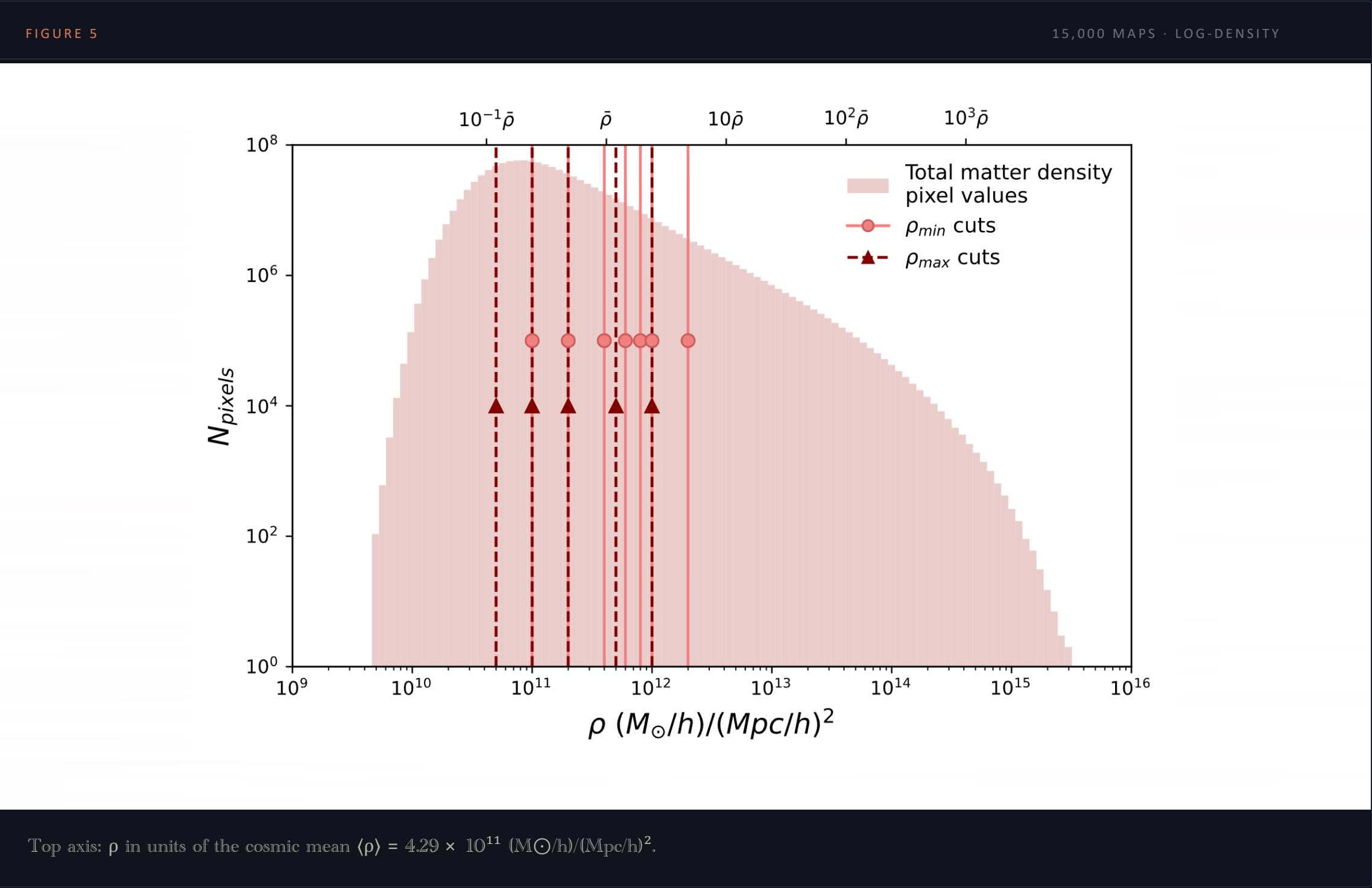
The low R^2 at $k \approx 1$ is partly a finite-box-size effect — for larger volumes the saturation knee would sit at lower k .

04 ----- DENSITY CUTS

Voids, filaments, or halos?

Erase low- ρ regions, then erase high- ρ regions, and watch what the network keeps using.

Figure 5 | pixel density histogram, with cut levels.



ROUGH TAXONOMY

VOIDS

$\rho < \langle \rho \rangle / 2$

FILAMENTS

$5 \langle \rho \rangle \lesssim \rho \lesssim 50 \langle \rho \rangle$

HALOS

$\rho \gtrsim 100 \langle \rho \rangle$

CUT METHOD

Replace each cut pixel with a log-uniform random value between the global min and max — effectively *delete* any structure in that regime.

Underdensity cuts | delete the voids.

For each threshold ρ_{min} , every pixel with $\rho < \rho_{\text{min}}$ is replaced by a log-uniform random value spanning the global pixel range. The network is then retrained from scratch.

CUTS SWEEP

$\rho_{\text{min}} = 1, 2, 4, 6, 8 \times 10^{11}$ and $10^{12}, 2 \times 10^{12}$
 $(\text{M}\odot/\text{h})/(\text{Mpc}/\text{h})^2$

IN UNITS OF MEAN DENSITY

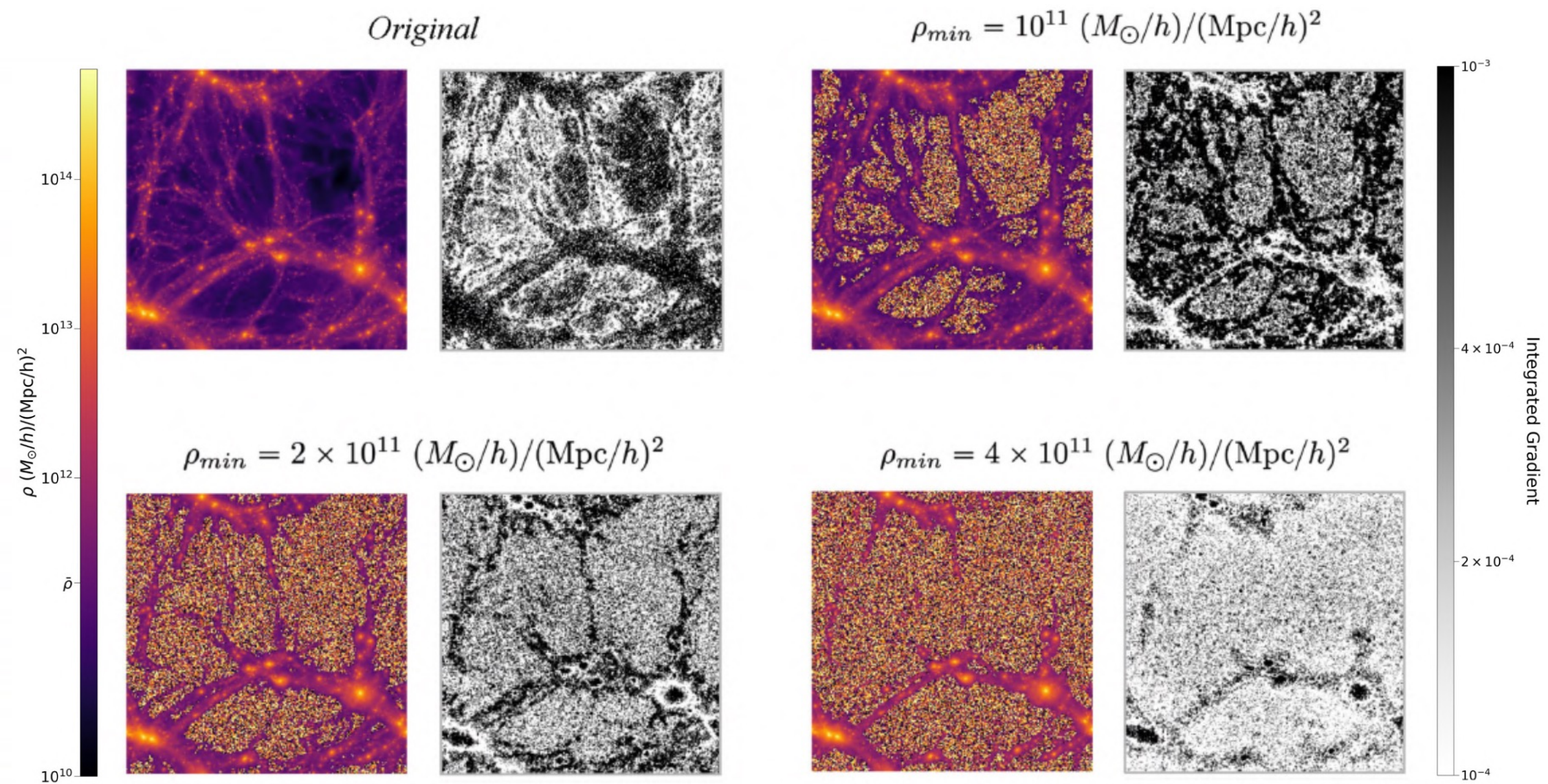
$\sim \langle \rho \rangle / 4$ up to $\sim 5 \langle \rho \rangle$

QUESTION

How much R^2 do we lose as we erase progressively shallower underdensities?

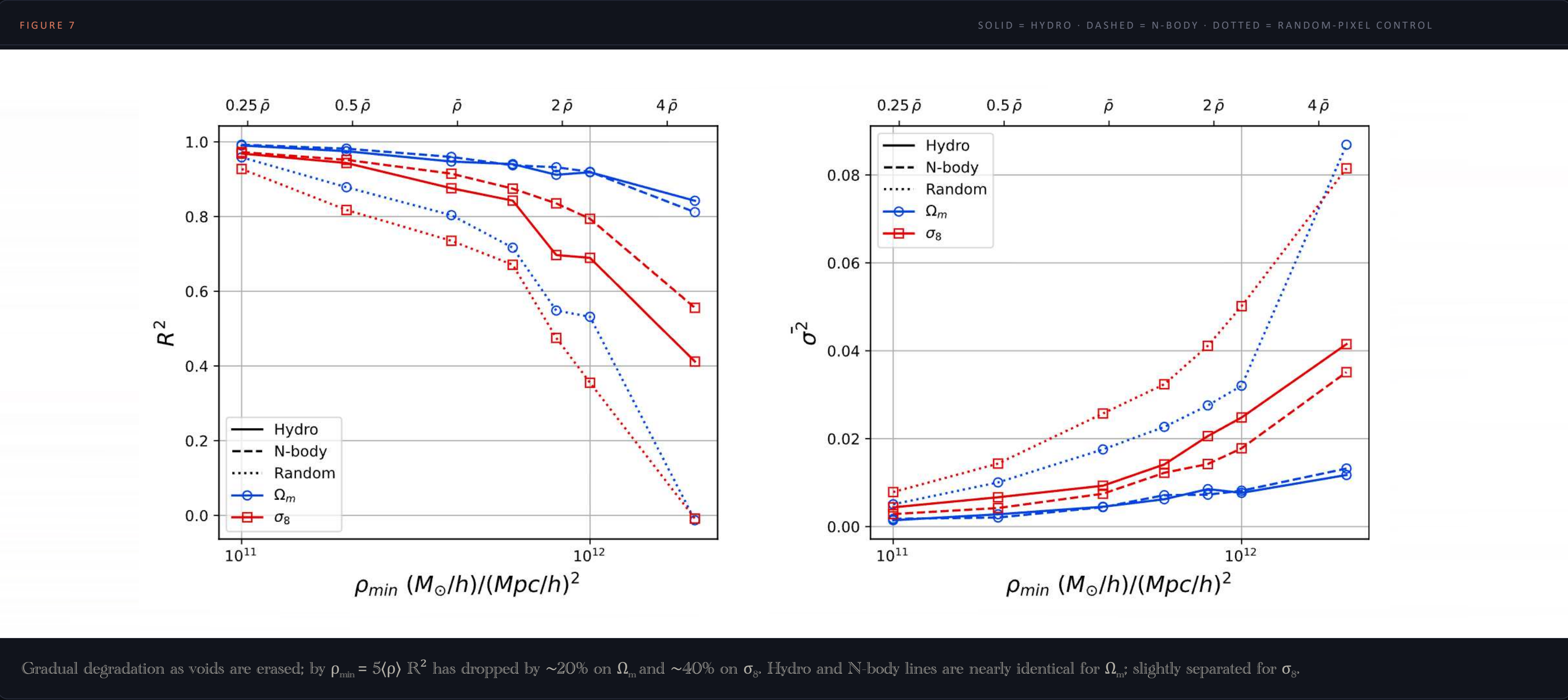
Figure 6 | IG maps shift away from voids.

FIGURE 6

LEFT: AUGMENTED MAPS · RIGHT: IG FOR $\Omega_M = 7$ VALUES OF P_{MIN} 

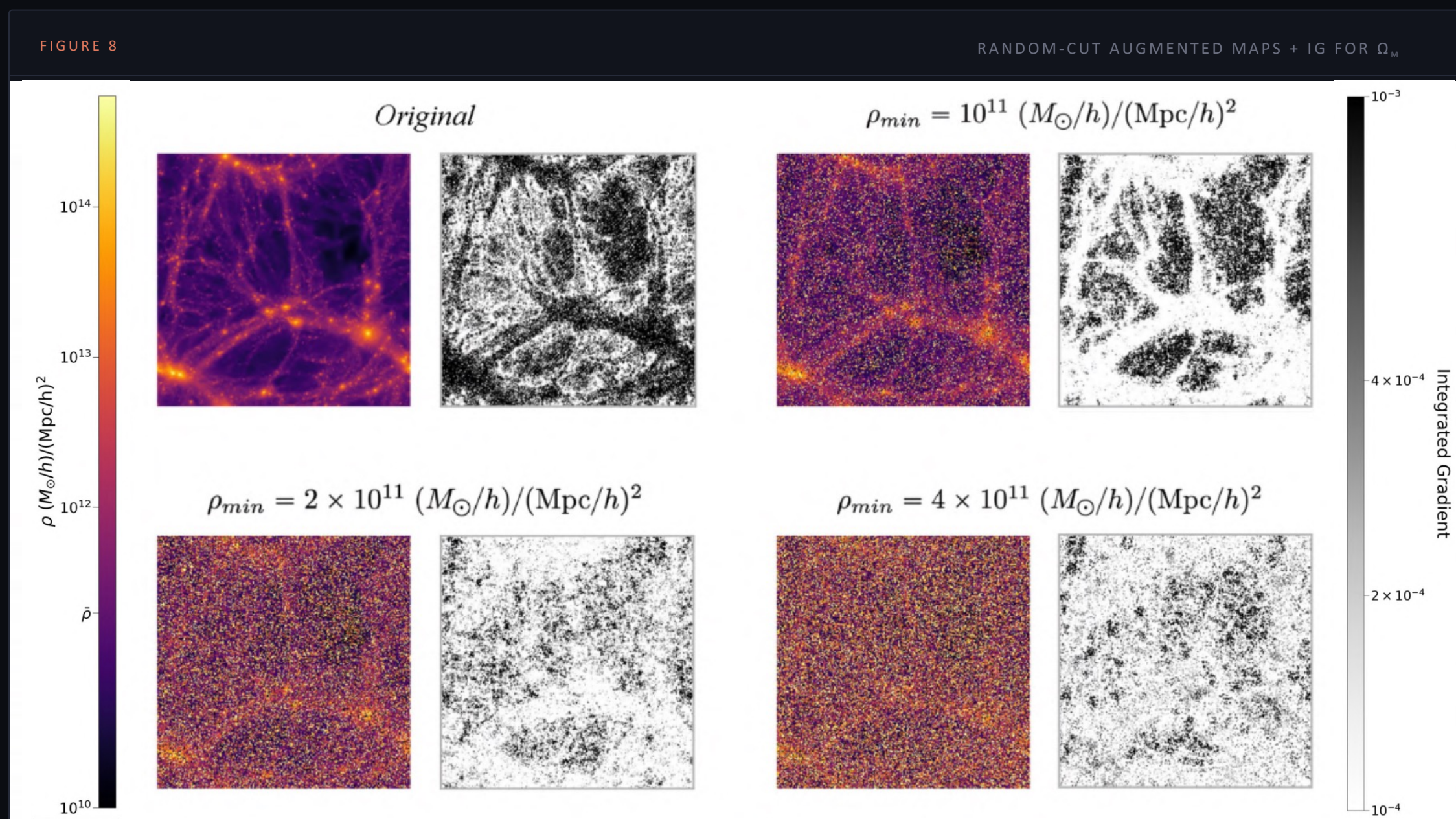
With no cut the IG lights up both voids and clusters. As ρ_{min} climbs, the void attribution vanishes — the network reorganises to extract information from filaments and halos.

Figure 7 | R^2 and $\langle \sigma^2 \rangle$ vs $\rho_{\min}$



Is it just pixel count? *No.*

Control experiment — randomise the **same number of pixels** as each $\rho_{\min}$ cut, but pick them *at random* instead of by density.



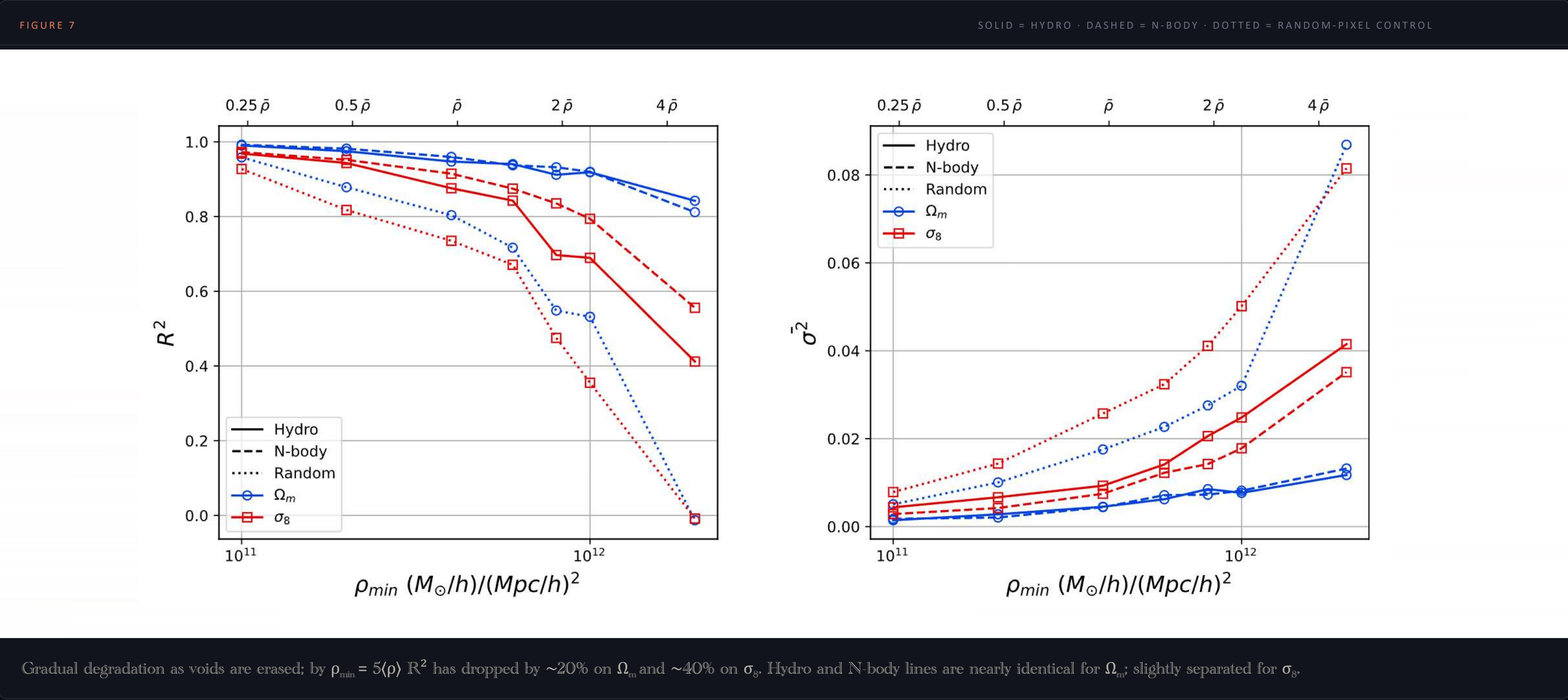
DOTTED LINES IN FIG. 7

R^2 is worse under random cutting than under density-based cutting at the same pixel budget.

INTERPRETATION

The CNN exploits the **spatially coherent structure** of voids — their connectedness and shape — not just their pixel count. If you have to throw away pixels, throw them away by density, not at random.

Figure 7 | R^2 and $\langle \sigma^2 \rangle$ vs $\rho_{\min}$



Overdensity cuts | delete the halos and filaments.

Now flip the procedure: replace every pixel with $\rho > \rho_{\text{max}}$ with a log-uniform random value. Sweep ρ_{max} from 10^{12} (halos only erased) all the way down to 5×10^{10} ($\sim \langle \rho \rangle / 10$ — only deep void interiors left).

WHY THIS MATTERS

Overdense regions are exactly where baryonic feedback is hardest to model. If the CNN can do without them, that's a robustness win.

Figure 9 | IG maps shift away from clusters.

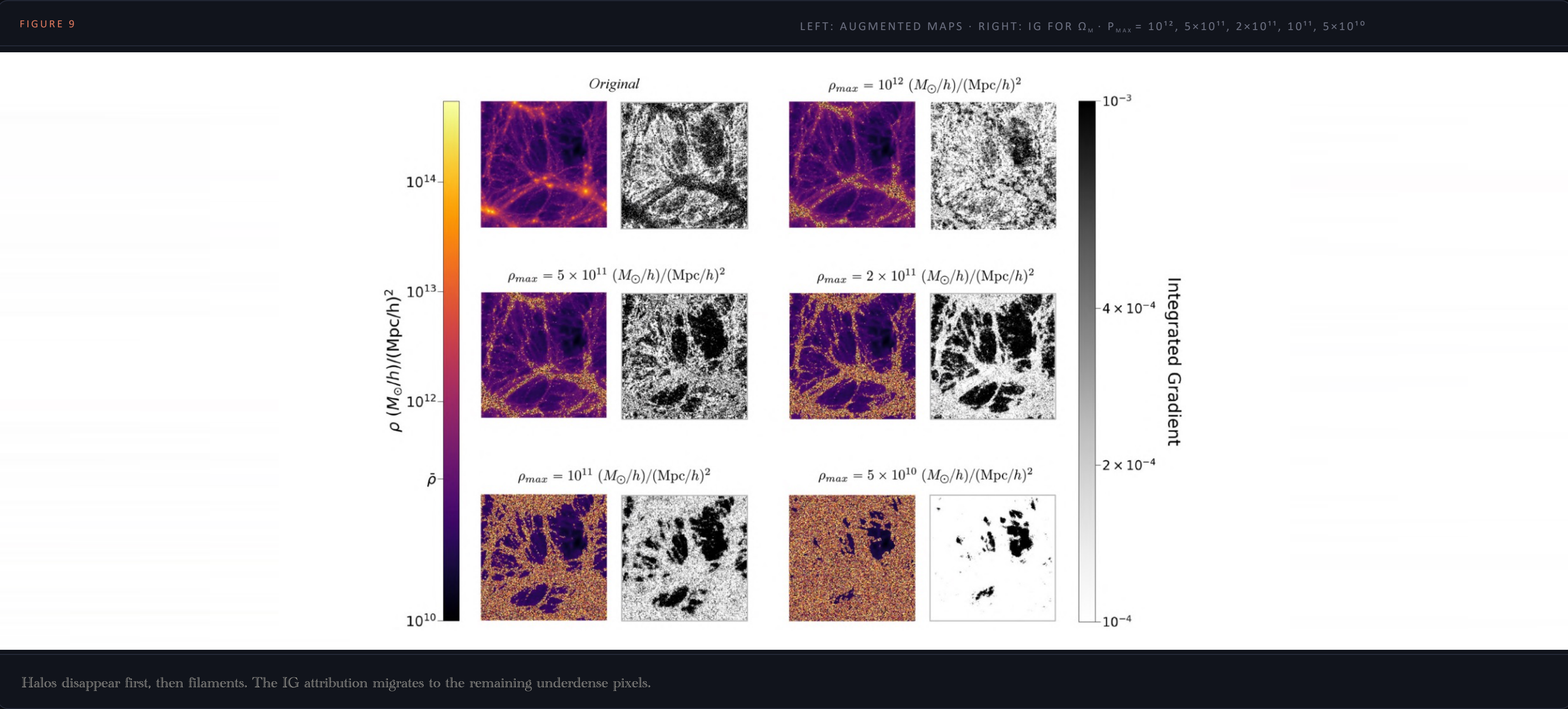
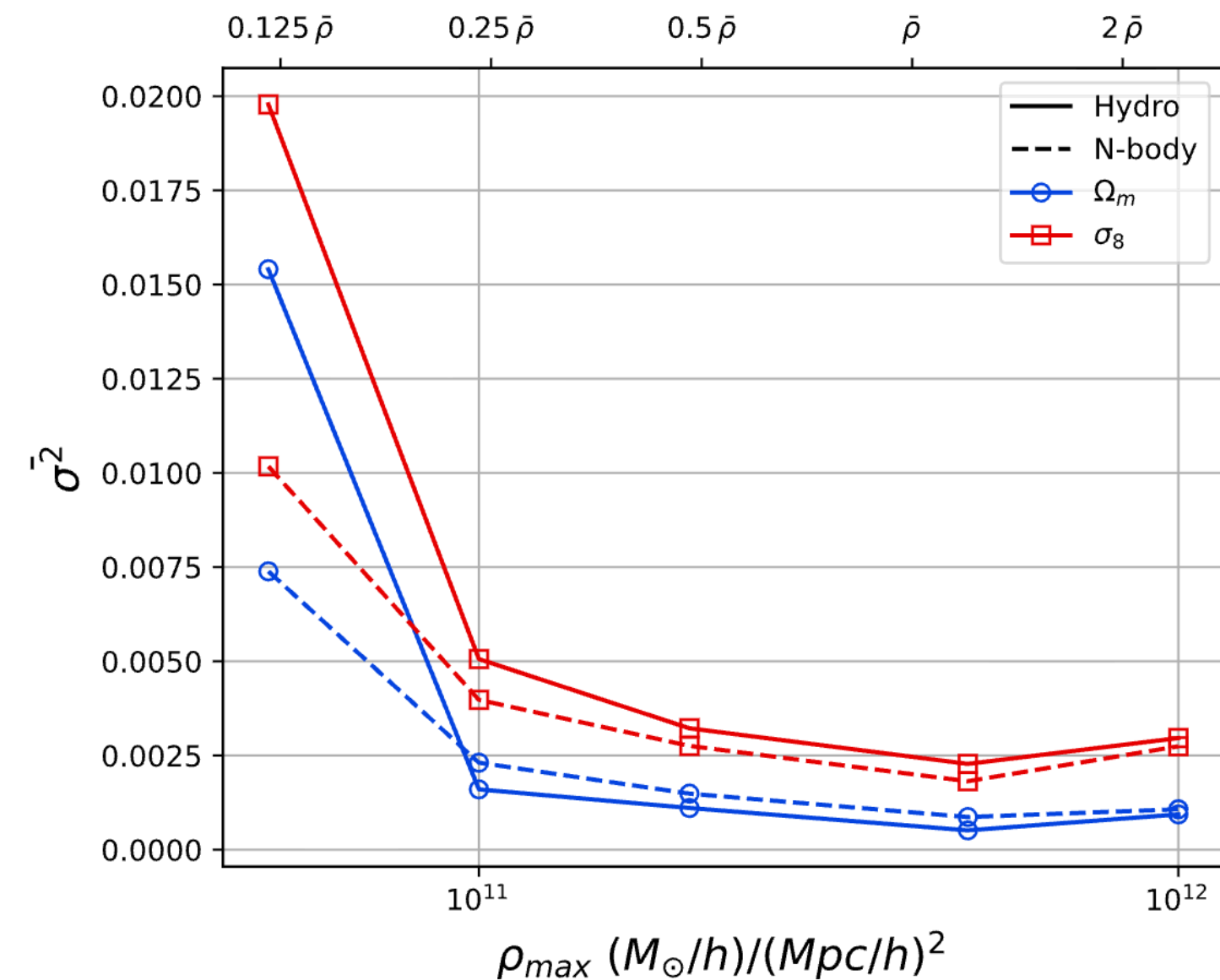
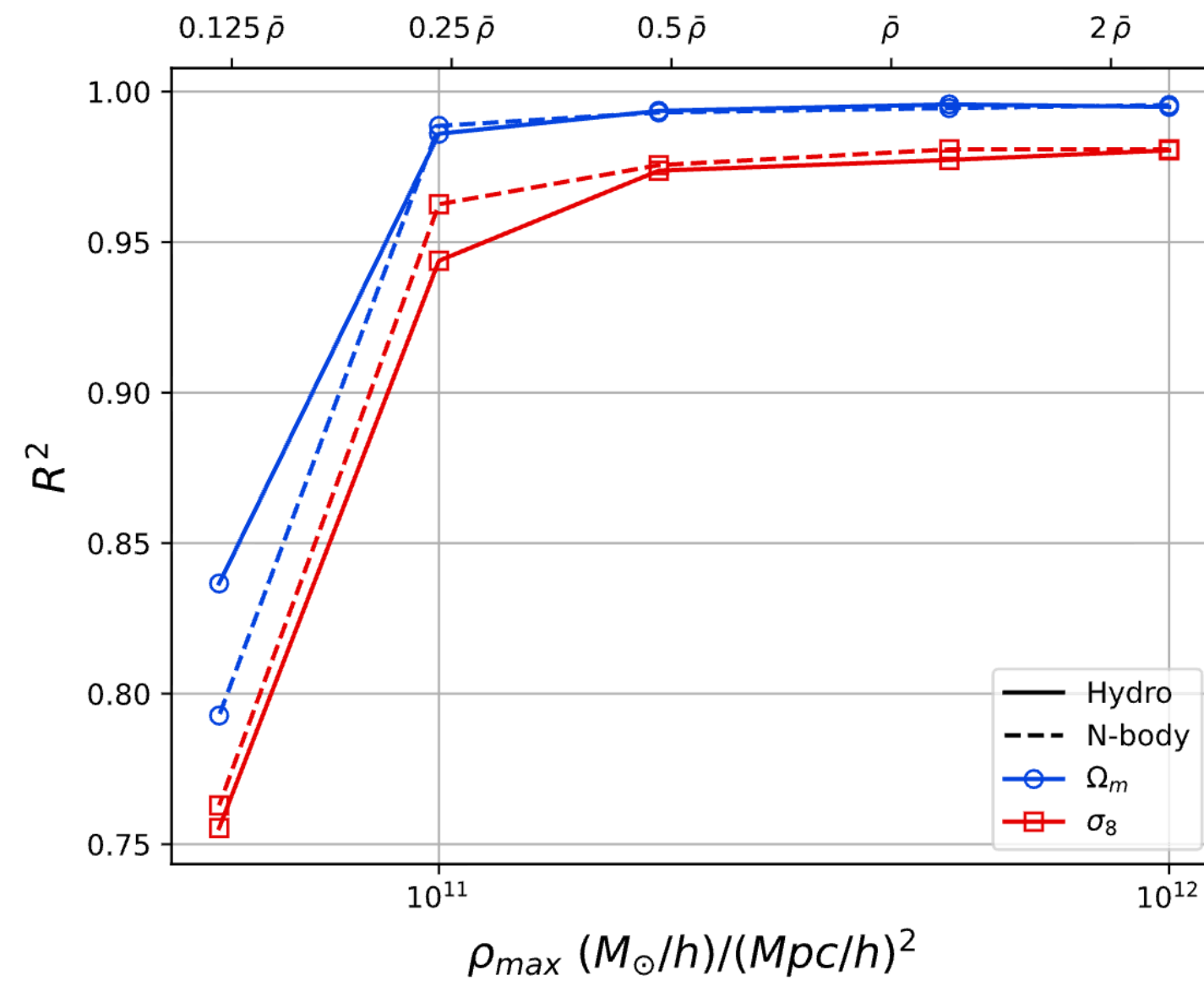


Figure 10 | inference saturates *deep inside the void regime*.

FIGURE 10

SOLID = HYDRO · DASHED = N-BODY

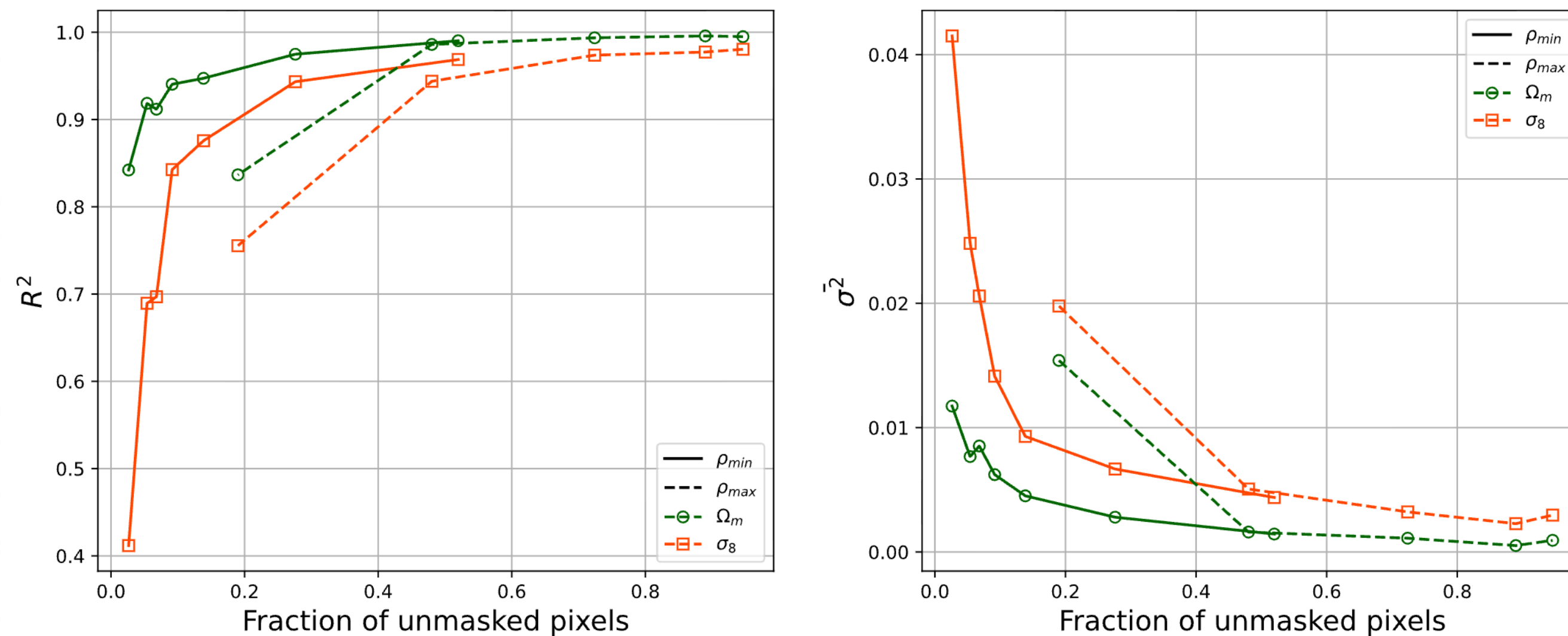


R^2 saturates already at $\rho_{\max} \approx \langle \rho \rangle / 4$ — well inside the void regime. The network learns almost everything about Ω_m and σ_8 from relatively underdense, void-like pixels alone.

But are voids actually informative *per pixel*?

Voids cover most of the area. If you re-plot R^2 against **fraction of unrandomised pixels** rather than threshold, the underdense and overdense curves split.

FIGURE 11

SOLID = $P_{\min}$ CUTS (KEEP DENSE PIXELS) · DASHED = $P_{\max}$ CUTS (KEEP VOID PIXELS)

The $\rho_{\min}$ (solid) curves sit at or above the $\rho_{\max}$ (dashed) curves — particularly for σ_8 . Overdense regions carry **more information per pixel**; voids only dominate because they're huge.

Are **baryons** messing this up?

Compare the solid (hydro) and dashed (N-body) lines in Figures 7 and 10. Same network architecture, same procedure — just trained on gravity-only maps that contain no astrophysical feedback.

FOR Ω_{M}

Indistinguishable.

Hydro and N-body curves overlap. The CNN has effectively learned to marginalise over feedback.

FOR Σ_{δ}

Slight degradation.

Baryonic suppression of small-scale power injects extra variance into σ_{δ} — but only mildly.

IMPLICATION

The CNN is reading the cold dark matter morphology. Baryons modulate but do not obscure the cosmological signal.

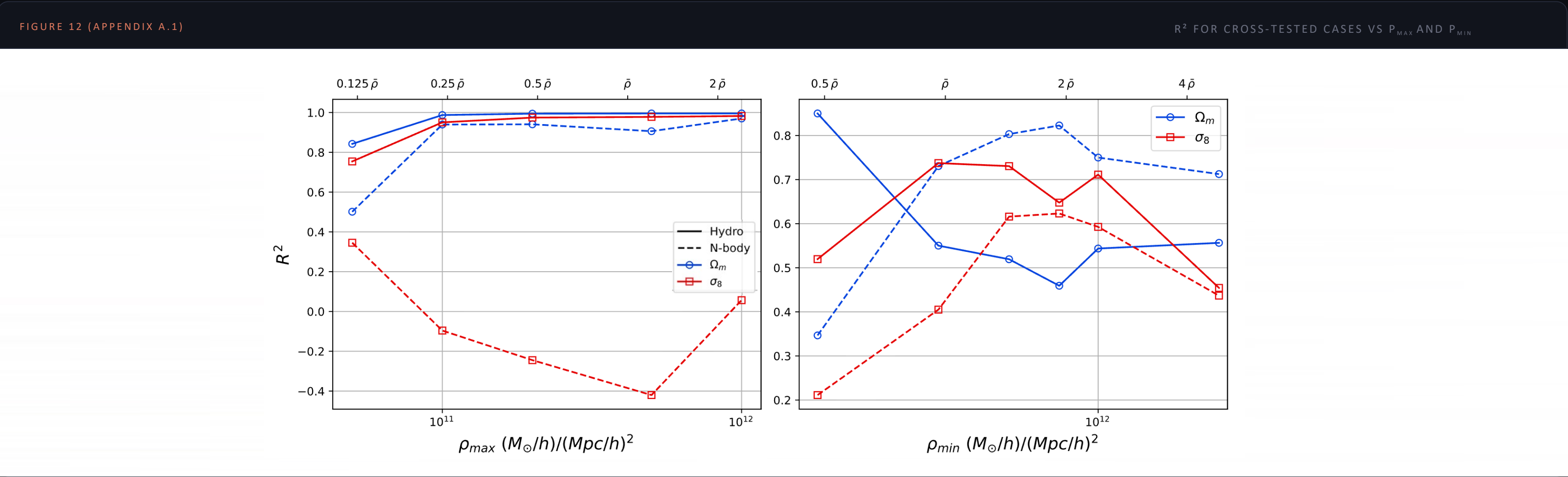
Cross-test | train on one, evaluate on the other.

TRAIN: HYDRO → TEST: N-BODY

Both Ω_m and σ_8 recovered with $R^2 \gtrsim 0.9$. The model has learned the underlying CDM morphology while marginalising over baryons.

TRAIN: N-BODY → TEST: HYDRO

Ω_m still fine. σ_8 collapses to $R^2 \approx 0$. A model that's never seen feedback misreads baryon-induced suppression as a shift in σ_8 .



Lesson: train on the more general dataset — the one that includes baryons.

Seven findings, and what they mean.

Implications for upcoming surveys · caveats · future directions.

Seven findings.

- 01

The CNN extracts cosmological information from **both deep voids and overdense halos** — the topological extremes of the cosmic web.
- 03

Removing underdense pixels (**$\rho_{\min}$ cuts**) gradually but significantly degrades inference — voids carry real information.
- 05

Density-based cutting **beats random cutting** at the same pixel budget — the network uses spatial coherence.
- 07

Per-pixel, **overdense regions carry more information than underdense ones**. Voids dominate in aggregate because of their volume and shape coherence — not because each void pixel is uniquely informative.

- 02

Cutting at **$k_{\max} = 10$ vs 50** costs only $\sim 10\%$ R^2 . Aggressive Fourier cuts come essentially for free.
- 04

Removing overdense pixels (**$\rho_{\max}$ cuts**) below $\langle \rho \rangle / 4$ changes inference very little — voids alone almost suffice.
- 06

Hydro vs N-body: Ω_m is unaffected; σ_8 takes a small hit. The CNN learns CDM morphology, marginalising over baryons.

Why should **observers** care?

Late-time probes — **galaxy clustering, weak lensing, 21 cm, SZ** — all suffer baryonic contamination on small scales.

Our paper says: you can cut hard in **k** and in **ρ** without sacrificing constraining power. Robustness $\approx$ free.

Density-based filtering is underexplored in cosmological analyses. Typical pipelines only cut in k.

Voids may be more powerful — relatively — for cosmological inference than current pipelines suggest. The complementarity of overdense and underdense regions is now confirmed at the field level.

OPTIMAL FILTERING — AN OPEN DIRECTION

*Some combination of **k**, **ρ**, and other field features could give robust, near-optimal cosmology from upcoming surveys.*

Caveats & open questions.

CAVEAT 01

Noise-free maps.

Real survey noise hits *underdense regions hardest* (Zorrilla Matilla 2020, Golshan & Bayer 2024). The per-pixel ranking may tilt further toward overdense regions in real data.

CAVEAT 02

One subgrid model.

Only the IllustrisTNG suite was used. Robustness *across* codes (e.g. SIMBA, Astrid) is the next test — flagged as future work by the authors.

CAVEAT 03

Small box.

25 h⁻¹ Mpc misses the largest modes. Sub-dividing larger sims (e.g. Quijote, Millennium) into 25-Mpc patches is a proposed cross-volume test.

Beyond these — there is room for non-axis-aligned, non-top-hat filters; combined **k+ρ** cuts; and applying the same interpretability framework to other observables (lensing, 21 cm directly).

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Even when the simulation is opaque, the network reading it is *not*.

The CNN attends to the extremes of the cosmic web — voids and clusters — extracts robust cosmology even in the presence of complicated baryonic physics, and tolerates aggressive cuts in both k and p . That's a hopeful result for the next generation of field-level cosmological inference.

DISCUSSION

Questions?

PAPER

Lahiry, Bayer & Villaescusa-Navarro 2025 ApJ 994,
129 · arXiv:2504.17839

RELEVANT PRIOR WORK

Villaescusa-Navarro+ 2021a (CAMELS) Villaescusa-
Navarro+ 2021b (CNN inference)

DATA

camel-simulations.org
CMD (Multifield Dataset)

Thank You



For Listening!